

$$\sum_{1 \leq i \leq n} a_i$$

$$\sum a_i = a_1 + a_2 + a_3 + \dots + a_n$$

$$\prod a_i = a_1 \times a_2 \times a_3 \times \dots \times a_n$$

$$\text{avg} = \left(\sum_{i=1}^{i=n} a_i \right) / n$$

$$\sum_{k=1}^n (2k+1) + \sum_{k=1}^n (2k-1)$$

$$\text{fact}(n) = \prod_{i=1}^{i=n} i$$

$$a + b \quad + \quad a \quad b \quad a \quad b +$$

$$a_1 + a_2 + a_3$$

$$\equiv (+ a_1 \ a_2) + a_3$$

$$\equiv + (+ a_1 \ a_2) \ a_3$$

$$\equiv (+ a_1 \ a_2 \ a_3)$$

$$a_1 + a_2 + a_3 + a_4$$

$$\equiv (+ a_1 \ a_2) + a_3 + a_4$$

$$\equiv (+ (+ a_1 \ a_2) \ a_3) + a_4$$

$$\equiv (+ (+ (+ a_1 \ a_2) \ a_3) \ a_4)$$

$$\equiv (+ a_1 \ a_2 \ a_3 \ a_4)$$

$$a_1 + a_2 + a_3 + \dots + a_n$$

$$\equiv (+ a_1 \ a_2 \ a_3 \ \dots \ a_n)$$

$$\equiv \left(+_{i=1}^{i=n} a_i \right)$$

+ Symbol

Summation

$$\equiv \left(\sum_{i=1}^{i=n} a_i \right)$$

$$\sum \quad \sum \quad \sum \quad \sum \quad \boxed{\sum}$$

$$\equiv \sum_{i=1}^{i=n} a_i$$

$$\begin{aligned} & a_1 \times a_2 \times a_3 \times a_4 \dots \times a_n \\ \equiv & (a_1 a_2 a_3 \dots a_n) \\ \equiv & \left(\prod_{i=1}^{i=n} a_i \right) \end{aligned}$$

$\times = \text{Product}$

$$\equiv \left(\prod_{i=1}^{i=n} a_i \right) \quad \gamma \quad p \quad p \quad \pi \quad \Pi$$

$$\equiv \left(\prod_{i=1}^{i=n} a_i \right)$$

$$\sum_{i=1}^{i=n} a_i = \sum_{1 \leq i \leq n} a_i \quad \Bigg| \quad \prod_{i=1}^{i=n} a_i = \prod_{1 \leq i \leq n} a_i$$

Inversion

$\{a_n\} = \text{Sequence}$

$$a_n = \boxed{n^2}$$

$$\sum_{k=1}^{k=50} a_k$$

$$\sum_{k=1}^{k=50} k^2$$

$$\sum_{1 \leq k \leq 50} k^2$$

$$n! = \prod_{1 \leq i \leq n} (i) \quad a_i = i$$

$$= \prod_{i=1}^{i=n} i$$

$$\equiv \prod_{i=1}^{i=n} i$$

$$\equiv (p \quad 1 \quad 2 \quad 3 \quad 4 \dots n) \\ \equiv (\times \quad 1 \quad 2 \quad 3 \quad 4 \dots n)$$

$$n! = (1 \times 2 \times 3 \times 4 \times \dots \times n)$$

- ① Perform common operation on every element in a container and return all results in the output container.
- ② Apply input condition on all elements in the input container, and return all elements for which cond was satisfied in the output container.

$$\{1-50\} \xrightarrow[\text{filter}]{\text{is_even}} \{2, 4, 6, \dots, 50\} \xrightarrow[\text{map}]{\text{Square}}$$

auto filter: [] (— , —) → std::vector<int>

auto map [] (— , —) → std::vector<int>

$$\sum_{\substack{1 \leq k \leq 50 \\ k \in \text{even}}} k^2$$

map([](int x) → int { return x * x; },
filter([](int y) → bool { return y % 2 == 0; }) (1, 50) }
 →
 std::vector<int> r =

`[] (std::vec`

```
[] (std::function<int(int)> func, std::vector<int>& vec) -> std::vector<int> {  
    std::vector<int> rsv;  
    for (auto& x: vec)  
        rsv.push_back(func(x));  
    return rsv  
}  
{  
    [] (int x) -> int { return x * x; },  
    [] (std::function<bool(int)> cond, std::vector<int>& vec) ->  
std::vector<int> {  
    std::vector<int> rsv;  
    for (auto& x : vec)  
        if (cond(x))  
            rsv.push_back(x);  
    return rsv;  
}  
} ( [] (int x) -> bool { return x % 2 == 0; }, std::vector<int> {1,2,3,4,5,6,  
7,8,9,10} ) )
```

```
result = reduce(  
    lambda x, y : x + y,  
    map(  
        lambda x ** 2,  
        filter(  
            lambda x : x % 2 == 0,  
            range(1, 11)  
        )  
    )  
)
```

```
result = 0  
for x in range(1, 11):  
    if x % 2 == 0:  
        result = result + x ** 2
```